

CBC Parameters

Adult Reference Intervals: Conventional vs SI

Parameter	Conventional Range	Conventional Units	SI Range	SI Units
WBC	4.5 – 11.0	$\times 10^3/\mu\text{L}$	4.5 – 11.0	$\times 10^9/\text{L}$
RBC	4.00 – 5.50	$\times 10^6/\mu\text{L}$	4.00 – 5.50	$\times 10^{12}/\text{L}$
HGB	12.0 – 17.4	g/dL	120 – 174	g/L
HCT	36.0 – 52.0	%	0.36 – 0.52	L/L
MCV	80.0 – 100.0	fL (or μm^3)	80.0 – 100.0	fL
MCH	28.0 – 34.0	pg	28.0 – 34.0	pg
MCHC	32.0 – 36.0	g/dL (or %)	320 – 360	g/L
RDW-CV	12.0 – 14.6	%	12.0 – 14.6	%
PLT	150 – 400	$\times 10^3/\mu\text{L}$	150 – 400	$\times 10^9/\text{L}$
MPV	6.8 – 10.2	fL	6.8 – 10.2	fL

Parameter	Conventional Range	Conventional Units	SI Range	SI Units
RBC (Male)	4.50 – 5.50	$\times 10^6/\mu\text{L}$	4.50 – 5.50	$\times 10^{12}/\text{L}$
RBC (Female)	4.00 – 5.00	$\times 10^6/\mu\text{L}$	4.00 – 5.00	$\times 10^{12}/\text{L}$
HGB (Male)	14.0 – 17.4	g/dL	140 – 174	g/L
HGB (Female)	12.0 – 16.0	g/dL	120 – 160	g/L
HCT (Male)	42.0 – 52.0	%	0.42 – 0.52	L/L
HCT (Female)	36.0 – 46.0	%	0.36 – 0.46	L/L

Critical (Panic) Values & Linearity

Critical Panic Values: Notify Physician



WBC <1.0 / >40.0



HGB <7.0 / >20.0



PLT <50 / $>1,000$ ($\times 10^3/\mu\text{L}$)

Analytical Linearity: Instrument Operational Range



WBC 1.0–99.9



HCT 10–60%



RBC 0.3–7.00



PLT 10–999



HGB 0.1–25.0

Exceeding Linearity: Dilute or Manual



Results marked “+++++” require a 1:2 dilution



Values below linearity require manual counts.

Derivation Formulas: RBC Indices

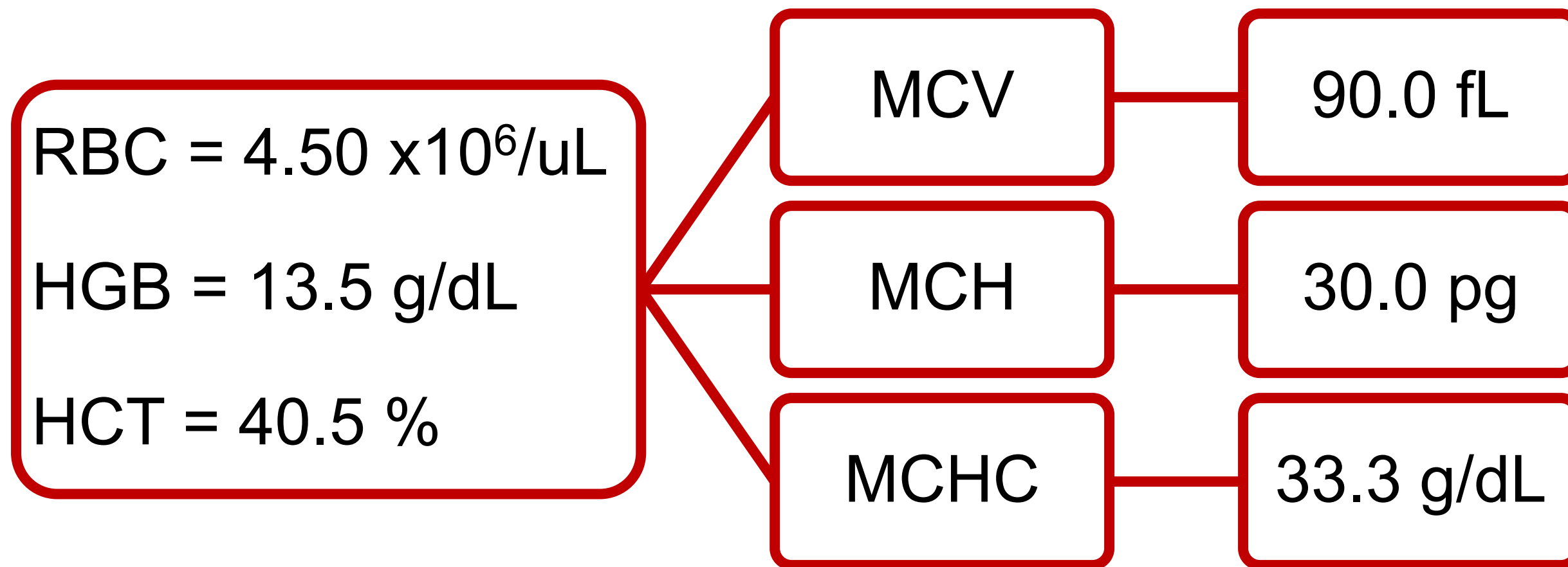
$$MCV = (HCT \times 10) / RBC$$

$$MCH = (HGB \times 10) / RBC$$

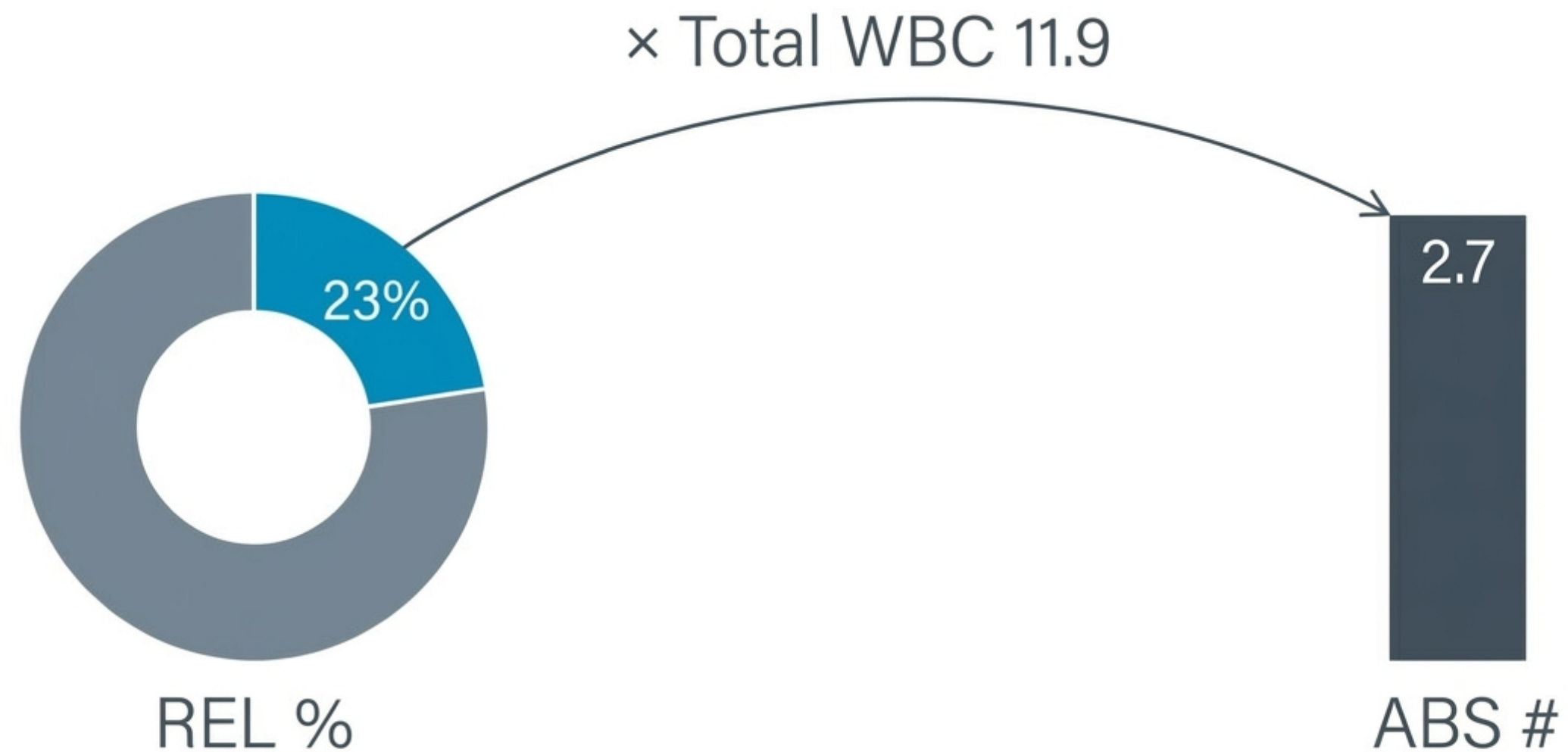
$$MCHC = (HGB \times 100) / HCT$$

$$RDW-CV = (MCV_{STD\ DEV} / MCV) \times 100$$

Worked Calculations: RBC Indices

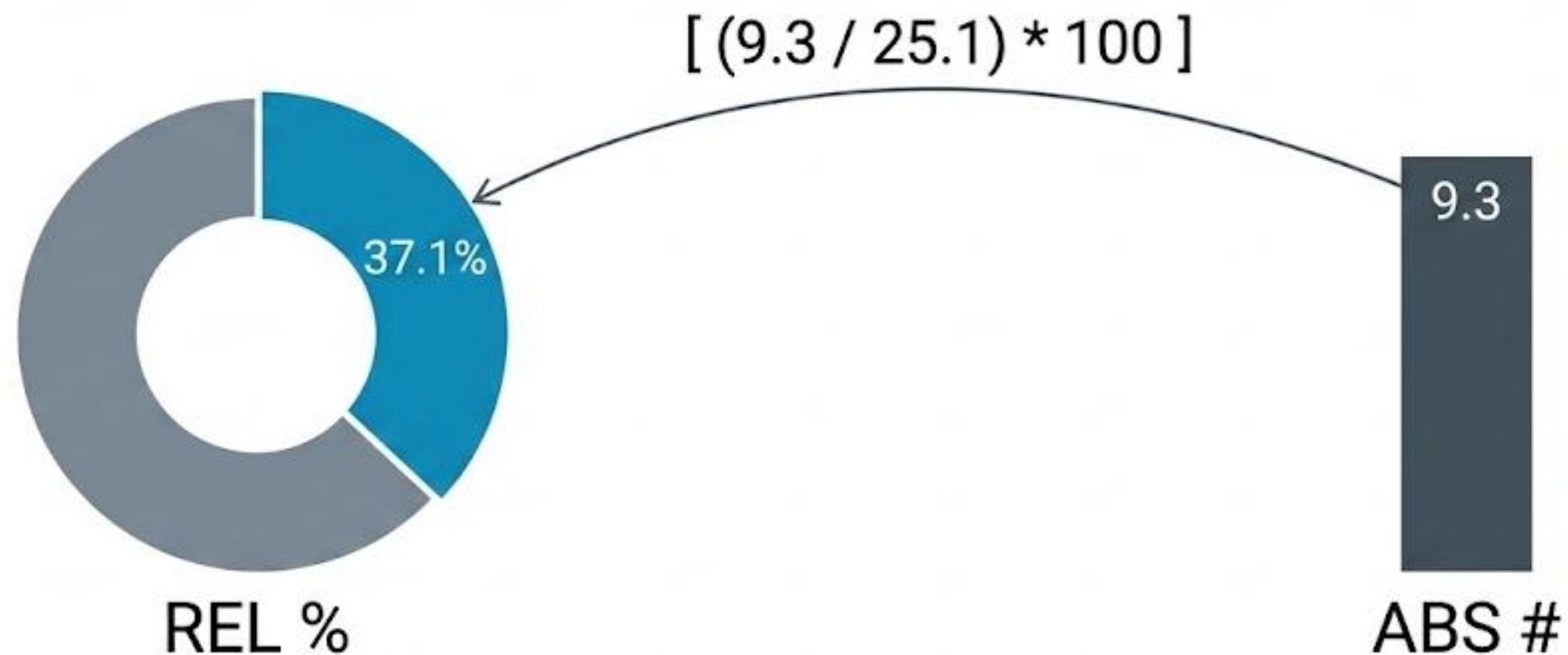


Worked Calculations: WBC Counts



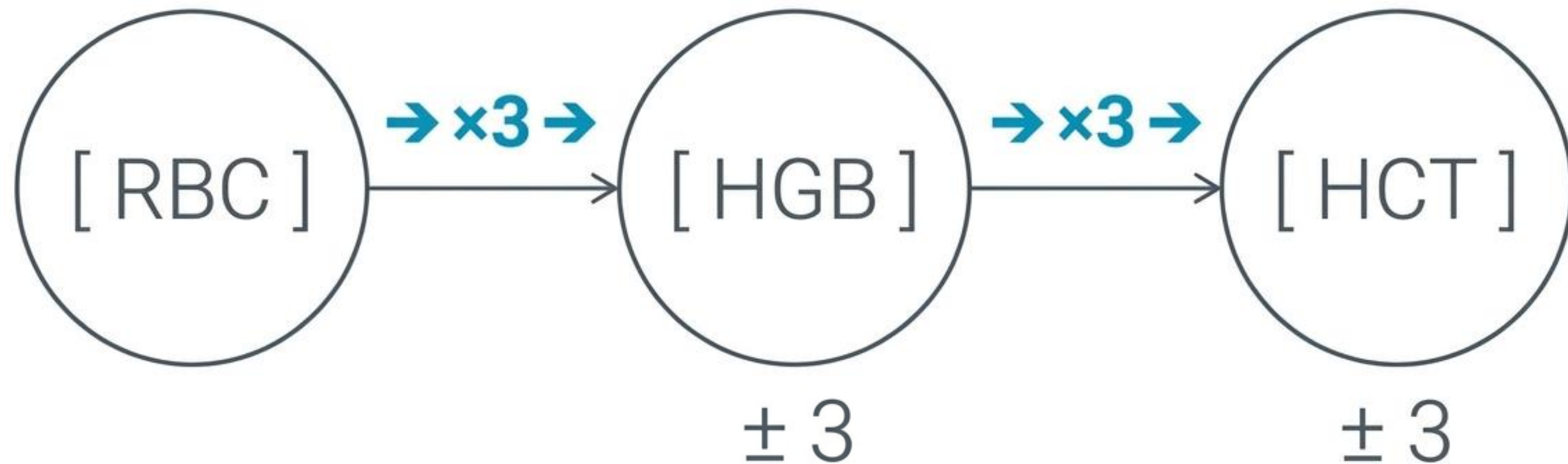
$$[\% \times \text{Total WBC} = \text{Abs Count}]$$

Worked Calculations: WBC Counts

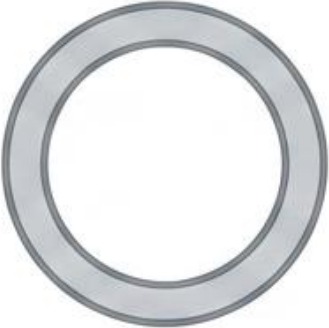


$$[(\text{Abs Count} / \text{Total WBC}) \times 100 = \text{Relative \%}]$$

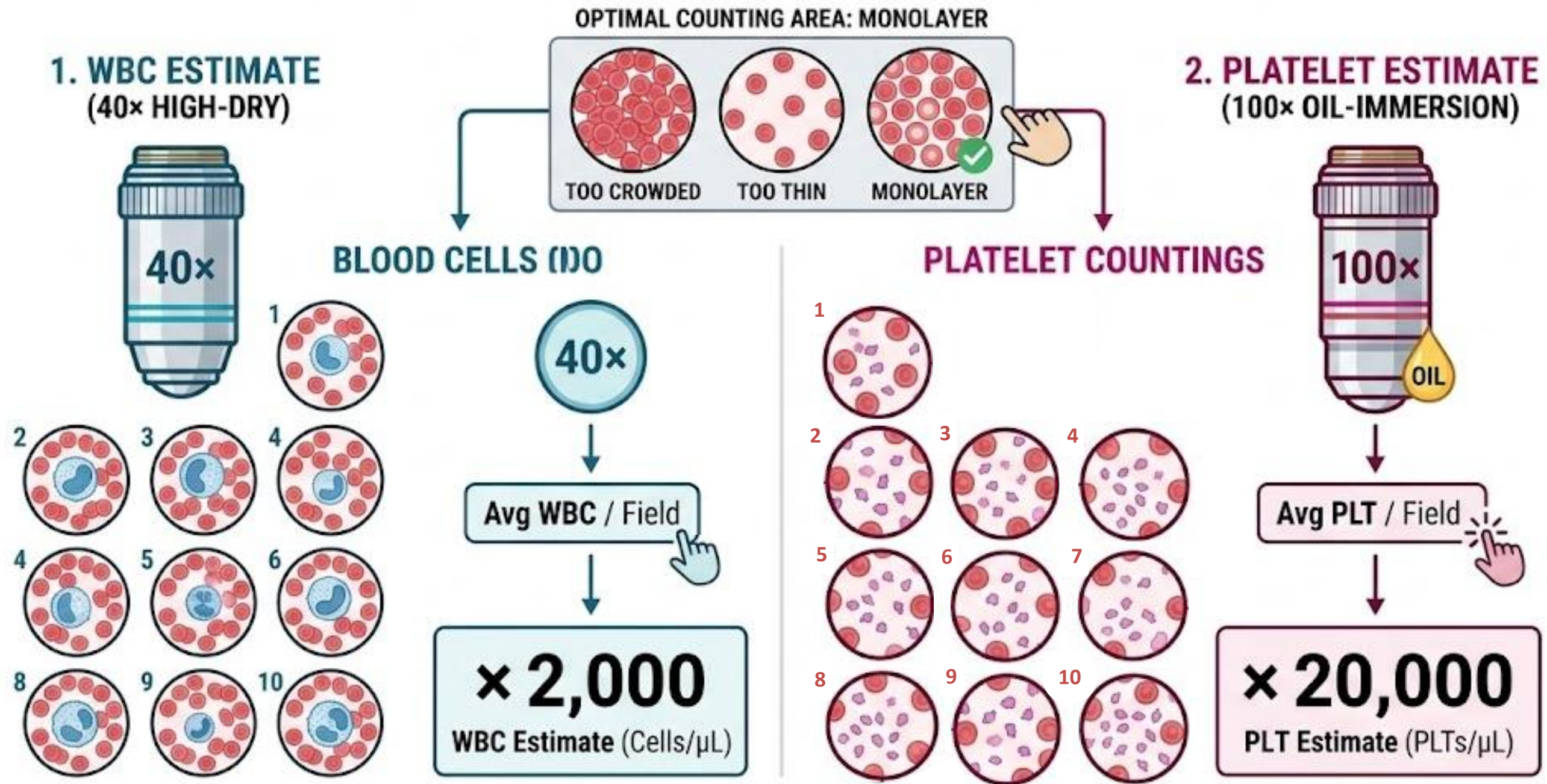
Internal QC Check: The Rule of Threes



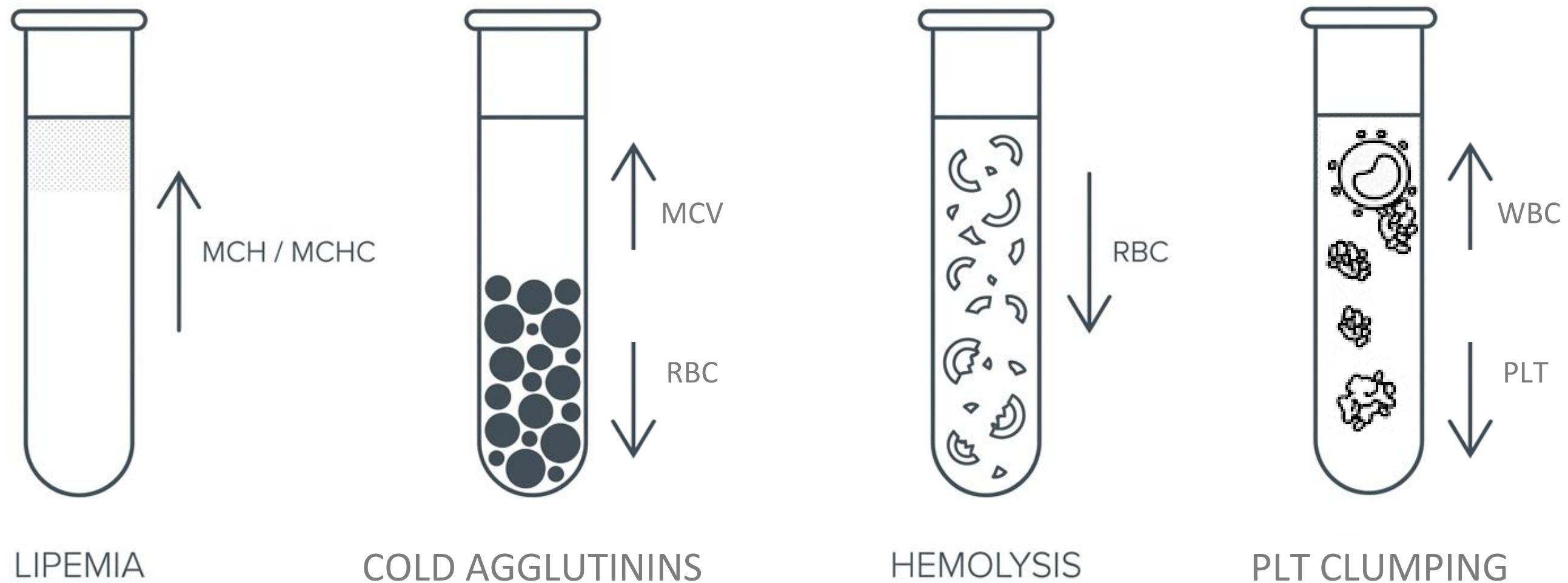
RBC Morphology Classification Matrix

		MCHC	
		< 32	32 - 36
MCV	> 100		
	80 - 100		
	< 80		

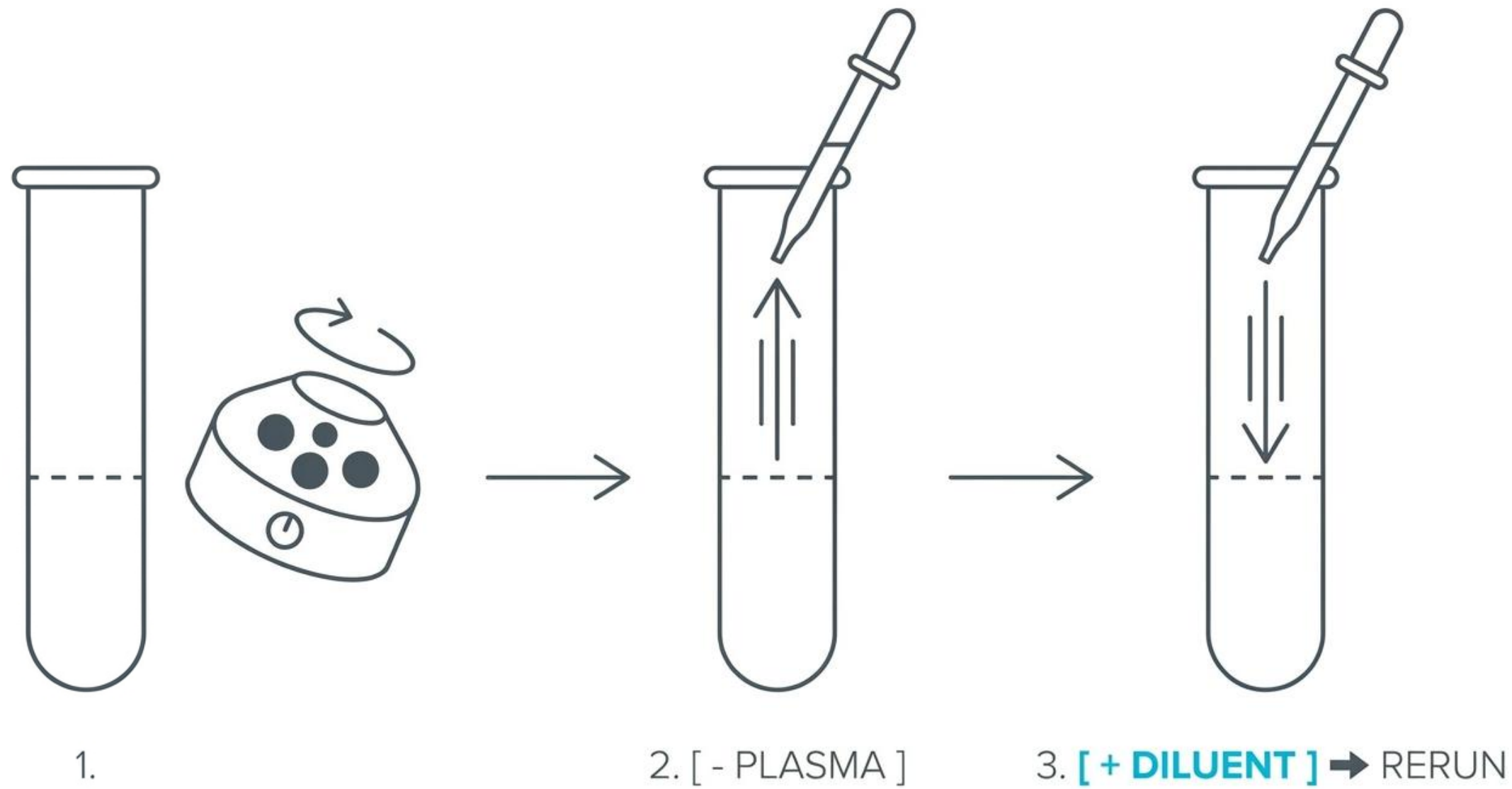
PBS Correlation & Slide Estimates



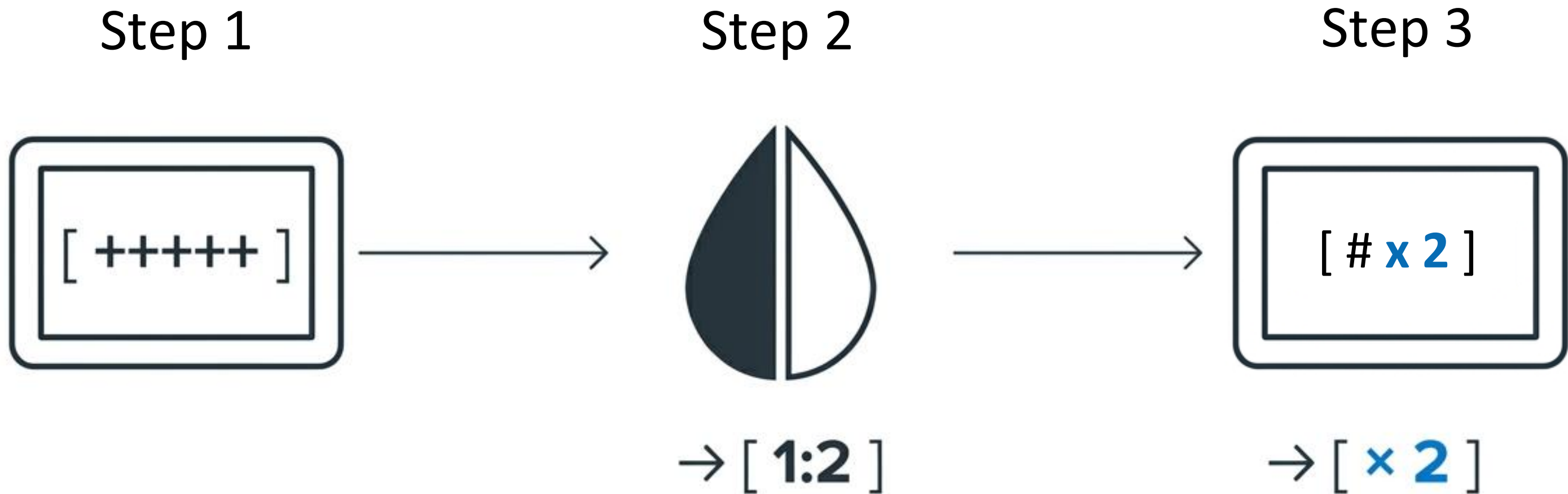
Automated CBC Interference Patterns



Plasma Replacement Protocol

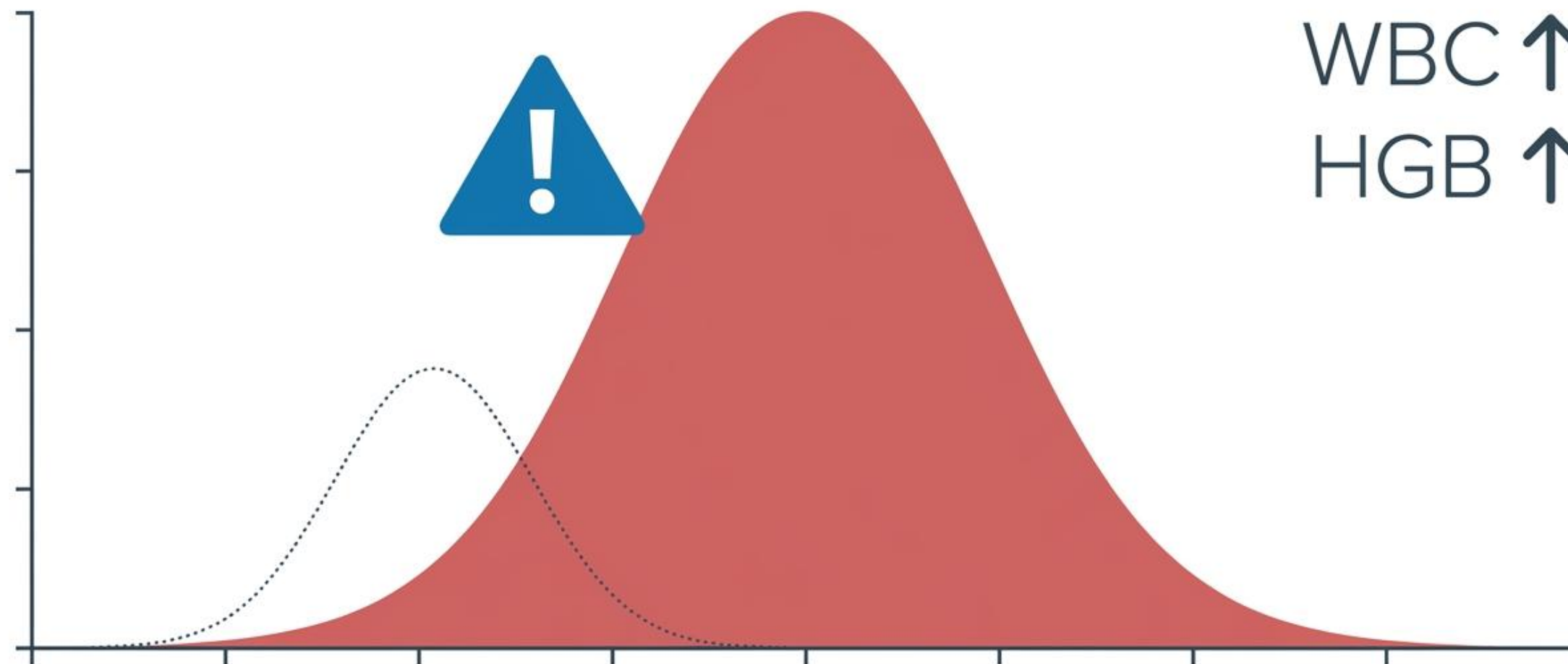


Out-of-Linearity Protocol



Failure Mode Analysis: Exhausted Lytic Reagent

NO LYSE



QC & Biosafety



DAILY QC



MAINTENANCE



PPE